

BUSINESS ANALYTICS

Capstone Topics 2026 for Postgraduate Class

Team size: 3–4 students | Project duration: 6 weeks

Focus on business decisions and measurable impact not model complexity alone

Dataset availability checked in September 2026

Format & Expectations

Each team will complete one capstone project using a clean and reproducible analytics pipeline. The project must begin with a business decision, not with a preferred algorithm. The final recommendation must be traceable from data, assumptions, analysis, and validation.

Required Project Package

- Reproducible code/notebook(s) with a README and clear run instructions.
- Data dictionary and source log: source URL, download date, filters, joins, and major cleaning decisions.
- Executive brief (maximum 2 pages) written for a manager or decision-maker.
- Small interactive dashboard focused on 3–5 decision-relevant KPIs and one actionable view.
- Model/analysis card summarizing target, features, validation design, assumptions, risks, ethics/bias considerations, and limitations.

Mandatory Analytics Rules

- Define the unit of analysis, target/outcome, prediction horizon (if any), and business decision before modeling.
- Use a validation design that matches the data-generating process. Time-dependent projects must use temporal holdout or rolling validation; random train/test splits are not sufficient when they create leakage.
- Use only information that would be available at the stated decision time. Explicitly identify and remove target leakage.
- Baseline is required. Stretch work is optional and should begin only after the baseline, validation, and business interpretation are complete.
- Do not claim causal uplift, ROI, avoided incidents, or cost savings unless the data support causal identification. Scenario-based estimates must state assumptions and include sensitivity analysis.
- Quantify business impact using an explicit decision rule, budget, threshold, capacity constraint, or scenario — not only predictive accuracy.

Suggested 6-Week Work Plan

Week	Primary focus	Expected checkpoint
1	Business framing, data acquisition, data dictionary, initial EDA	Problem statement + data readiness review
2	Cleaning, feature design, validation plan, baseline model/analysis	Reproducible baseline
3	Advanced model/analysis and error analysis	Model comparison + diagnostics
4	Decision rules, scenario simulation, business impact analysis	Actionable decision logic
5	Robustness/sensitivity checks, dashboard, executive interpretation	Draft dashboard + recommendations
6	Final QA, reproducibility check, executive brief, presentation/demo preparation	Final submission package

1. Customer Value & Promotion Targeting for an Online Retailer

Business Question

Which customers should the retailer prioritize for retention or promotion to increase expected future margin without over-discounting?

Primary Dataset

- UCI Machine Learning Repository – Online Retail II. Two years of transactional data for a UK-based non-store retailer; suitable for customer-level RFM, repeat-purchase, and value modeling. UCI Online Retail II
- Recommended scope: use the full dataset or a clearly documented time window; define an observation window and a future outcome window.

Core Tasks

- Clean transactions, identify cancellations/returns, and construct customer-level behavioral features.
- Build RFM/cohort views and a customer value segmentation.
- Predict repeat purchase and/or future customer value over a defined holdout horizon.
- Design a promotion-targeting rule and simulate expected margin under explicit response-rate and discount assumptions.

Baseline (Required)

- RFM segmentation plus one interpretable predictive model (e.g., logistic regression for repeat purchase and/or simple CLV/future-spend regression).
- Temporal holdout evaluation: features must be computed only from the observation window.
- Promotion scenario table using at least three discount/response assumptions.

Stretch (Optional / Bonus)

- Gradient boosting with calibration and SHAP/feature attribution.
- Budget-constrained targeting optimization or robust sensitivity analysis across response assumptions.
- Customer-level recommendation layer (e.g., retain / nurture / no-offer) with business rationale.

Success Metrics

- Repeat-purchase model: PR-AUC/ROC-AUC plus calibration.
- Future value/spend model: MAE/RMSE (preferably on a holdout period).
- Business metric: expected incremental margin or value under the simulated targeting policy versus a simple non-targeted policy.

Deliverables

- Customer analytics notebook/pipeline and customer-level score table.
- Executive brief: target segments, recommended policy, expected impact, and key assumptions.
- Dashboard: segment size/value, retention risk, and promotion scenario outcomes.

Scope Guardrails

- The dataset does not contain randomized promotion treatment/control. Do not report causal uplift or observed “revenue lift vs. control.”
- Promotion effects are scenario assumptions unless the team adds a separate experimental dataset with valid treatment information.
- Do not merge unrelated complaint datasets solely to increase feature count.

2. Bike-Share Demand Forecasting & Imbalance Management

Business Question

Can station-level demand be forecast well enough to identify stations at high risk of imbalance and support a practical rebalancing policy?

Primary Dataset

- Citi Bike NYC – System Data / Trip Histories. Public trip-level data with timestamps and start/end stations. [Citi Bike System Data](#)
- Recommended scope: 3–6 months of data and approximately 30–50 high-volume stations to keep feature engineering and simulation manageable.

Optional Enrichment

- Weather data may be added if the team can merge it reproducibly and without spending disproportionate effort on data acquisition.

Core Tasks

- Aggregate trip histories into hourly departures, arrivals, and net demand by station.
- Create calendar, lag, rolling, and station-level demand features.
- Forecast short-horizon demand and convert forecasts into an imbalance-risk score.
- Simulate a simple rebalancing policy using assumed station capacity/starting inventory.

Baseline (Required)

- Seasonal-naïve baseline plus one supervised model (e.g., XGBoost/LightGBM) or SARIMAX.
- Temporal validation with at least one true future holdout period.
- Simple rule-based rebalancing simulator driven by predicted net demand.

Stretch (Optional / Bonus)

- Weather enrichment and/or spatial neighboring-station features.
- Greedy or OR-Tools vehicle-routing heuristic for a limited subset of stations.
- Quantile forecasts for shortage-risk planning.

Success Metrics

- Demand forecasting: MAE/RMSE or WAPE for departures, arrivals, and/or net demand.
- Risk metric: recall/precision for high-imbalance station-hours under a defined threshold.
- Simulation metric: reduction in simulated shortage/overflow events and rebalancing distance/cost.

Deliverables

- Forecasting pipeline with station-hour predictions and risk scores.
- Rebalancing simulator with a clearly documented inventory/capacity assumption.
- Map dashboard showing demand, risk, and suggested rebalancing actions.

Scope Guardrails

- Trip histories do not provide complete historical station inventory. Treat shortage/overflow outcomes as simulated unless an appropriate inventory feed/history is added.
- A full city-scale vehicle-routing problem is not required. Keep optimization secondary to forecasting and decision quality.
- Do not use future trip counts to construct predictors for the same forecast horizon.

3. Airbnb Pricing & Demand-Proxy Analytics for Host Strategy

Business Question

Which listing characteristics are associated with price and demand proxies, and what practical host actions could improve expected revenue under transparent assumptions?

Primary Dataset

- Inside Airbnb – city-level listings, calendar, and reviews data. Choose one city and one clearly dated data snapshot. [Inside Airbnb – Get the Data](#)
- Recommended scope: one city only; use listings plus at most one additional table (calendar or reviews) unless the integration is straightforward.

Core Tasks

- Clean price and listing attributes; engineer interpretable features for room type, location, amenities, review activity, and availability.
- Model nightly price and analyze demand-related proxies such as review activity and calendar availability.
- Segment listings/hosts and identify actionable differences among comparable properties.
- Run “what-if” host scenarios (e.g., price change, amenity addition, minimum-stay change) with explicit assumptions.

Baseline (Required)

- Regularized regression for log-price plus descriptive/segmentation analysis for demand proxies.
- Geographic and property-type controls to avoid misleading cross-neighborhood comparisons.
- At least three host scenarios with sensitivity analysis.

Stretch (Optional / Bonus)

- Gradient boosting for price with SHAP/partial dependence.
- Comparable-listing matching or nearest-neighbor benchmark for pricing recommendations.
- Text features from listing names/descriptions if they add measurable value and remain interpretable.

Success Metrics

- Price model: MAE/RMSE (preferably on log-price and/or original currency with clear interpretation).
- Demand-proxy analysis: out-of-sample metric if modeled, otherwise stability/segment separation and robustness checks.
- Business metric: scenario-based expected revenue change, explicitly labeled as an estimate rather than causal impact.

Deliverables

- City-level analytical dataset and reproducible notebook/pipeline.
- Host playbook with 3–5 evidence-based recommendations and stated assumptions.
- Interactive map/dashboard showing price, demand proxies, and comparable listings.

Scope Guardrails

- Inside Airbnb calendar availability is not a ground-truth occupancy measure; blocked dates and booked dates cannot always be distinguished.
- Do not claim causal effects of host actions from observational cross-sectional data.
- Do not rely on ad-hoc Google search results as a required event-calendar data source.

4. Food Safety Risk Scoring for Inspection Prioritization

Business Question

If inspection capacity is limited, which establishments should be prioritized to maximize the number of high-risk inspections detected?

Primary Dataset

- City of Chicago – Food Inspections. Inspection records from 2010 to the present, including establishment, inspection type, result, and related fields. [Chicago Food Inspections](#)
- Recommended scope: use a recent multi-year window and predict the outcome of a future inspection using only information available beforehand.

Core Tasks

- Create establishment history features (prior failures, time since last inspection, inspection frequency, facility type, geography).
- Define a high-risk target (e.g., failed inspection) and inspect class imbalance.
- Build and calibrate a risk score.
- Evaluate inspection prioritization under a fixed inspection budget (e.g., top 5%, 10%, 20%).
- Perform a limited fairness/coverage analysis across geography and/or facility categories.

Baseline (Required)

- Logistic regression plus one tree-based model.
- Temporal split by inspection date and strict leakage audit.
- Top-k targeting analysis showing how many failed inspections would be captured under different capacity levels.

Stretch (Optional / Bonus)

- Calibrated gradient boosting and cost-sensitive learning.
- Spatial features based on prior nearby inspection outcomes.
- Alternative threshold policies balancing recall, capacity, and subgroup coverage.

Success Metrics

- PR-AUC and recall@k/top-k capture rate for failed/high-risk inspections.
- Calibration (e.g., Brier score and reliability plot).
- Business/operations metric: expected number of high-risk inspections identified under a fixed inspection budget.

Deliverables

- Risk-scoring pipeline and prioritized establishment list.
- Inspector dashboard with risk distribution, map, and capacity slider/threshold view.
- Short policy memo explaining the targeting rule, fairness checks, and operational limitations.

Scope Guardrails

- Features from the inspection being predicted (especially Results and contemporaneous Violations) are target leakage and must not be used as predictors.
- Do not claim “avoided incidents” unless a separate public-health outcome dataset and a defensible causal design are provided.
- Duplicate or repeat inspections must be handled carefully and documented.

5. Flight Delay Risk & Cost Scenario Analytics

Business Question

Can departure delay risk be predicted early enough to prioritize operational attention and reduce expected delay-related cost under a transparent cost model?

Primary Dataset

- U.S. DOT Bureau of Transportation Statistics – Reporting Carrier On-Time Performance. [BTS On-Time Performance](#)
- Recommended scope: one to three airports, carriers, or major routes over 3–6 months rather than the entire U.S. system.

Core Tasks

- Define a prediction time (e.g., two hours before scheduled departure).
- Engineer schedule, route, carrier, calendar, and historical congestion/delay feature available at that time.
- Model delay occurrence (>15 minutes) and/or delay minutes.
- Translate predicted risk into an operational prioritization rule and a scenario-based expected-cost calculation.

Baseline (Required)

- Logistic regression or Random Forest/LightGBM for delay classification, plus a simple regression if delay minutes are modeled.
- Temporal holdout and leakage audit based on the stated prediction time.
- Cost scenario with at least three plausible per-delay cost assumptions.

Stretch (Optional / Bonus)

- Weather enrichment from a reproducible source.
- Calibrated gradient boosting plus SHAP/feature attribution.
- Hub-level congestion proxy using rolling inbound/outbound traffic available before prediction time.

Success Metrics

- Classification: PR-AUC/ROC-AUC, recall for >15-minute delays, and calibration.
- Regression: MAE for delay minutes.
- Business metric: expected cost under the model-based prioritization policy versus a simple baseline policy, reported as scenario results.

Deliverables

- Reproducible delay-risk dataset and prediction notebook/pipeline.
- Operations brief with high-risk periods/routes and recommended prioritization rules.
- Dashboard showing delay risk, drivers, threshold/capacity trade-offs, and cost scenarios.

Scope Guardrails

- Actual departure time, realized taxi-out time, realized delay causes, and other post-decision variables must not be used if they are unavailable at prediction time.
- Crew, voucher, and missed-connection costs are not fully observed in BTS data. Treat them as scenario parameters unless an additional source is introduced.
- A deployed API is not required for the six-week project.

6. Road Collision Risk Mapping & Intervention Prioritization

Business Question

Where and when are severe road collisions concentrated, which factors are associated with severity, and how should a limited intervention budget be prioritized?

Primary Dataset

- Choose ONE jurisdiction only: NYC Open Data – Motor Vehicle Collisions (Crashes), or UK Department for Transport – Road Safety (STATS19). [NYC Motor Vehicle Collisions](#)
- UK alternative: STATS19 road safety open data. [UK Road Safety Open Data](#)
- Recommended scope: one jurisdiction and approximately 2–3 recent years, or a smaller geographic subarea if needed.

Core Tasks

- Perform spatial and temporal EDA and construct hotspot/risk units (grid cells, intersections, or road segments where feasible).
- Detect persistent hotspots using a transparent method (e.g., KDE, DBSCAN, or ranked collision density).
- Model severe injury/fatal collision risk using pre-outcome context features.
- Prioritize locations under a fixed intervention budget and propose evidence-aligned countermeasure categories.

Baseline (Required)

- Hotspot map plus logistic/ordinal severity model.
- Temporal or spatially aware validation; document how repeated locations are handled.
- Budget-constrained prioritization list (e.g., top 20 locations/zones).

Stretch (Optional / Bonus)

- Gradient boosting with spatial/context features.
- Hotspot stability analysis across time periods.
- OpenStreetMap road-context enrichment if reproducibly obtained.

Success Metrics

- Severity model: PR-AUC and recall@k for severe collisions.
- Hotspot quality: concentration of severe collisions captured in top-ranked areas and stability over time.
- Decision metric: share of severe-collision burden covered by a fixed number/budget of prioritized locations.

Deliverables

- Reproducible geospatial pipeline and risk/hotspot layer.
- Interactive map dashboard with time, severity, and intervention-priority views.
- Policy brief with prioritized locations, recommended intervention categories, assumptions, and limitations.

Scope Guardrails

- Do not combine NYC and UK datasets in one six-week project.
- Collision data alone does not identify the causal effect or ROI of a specific road intervention. Any cost-benefit result must be scenario-based and clearly labeled.
- Avoid random splitting that places nearly identical to nearby/repeated locations in both training and test sets without justification.

7. Profit-Aware Telecom Churn & Retention Targeting

Business Question

Given a limited retention budget, which customers should receive an offer when both churn risk and expected economic value are considered?

Primary Dataset

- IBM Telco Customer Churn sample dataset (commonly distributed through Kaggle). The dataset represents a fictional telco and is appropriate for teaching churn analytics. [Kaggle – Telco Customer Churn](#)
- The team may create a transparent synthetic offer-cost/retention-response table for business simulation; assumptions must be documented.

Core Tasks

- Perform churn EDA and identify customer segments with different risk/value profiles.
- Build and calibrate a churn probability model.
- Construct a simple customer value proxy using tenure, charges, and remaining-value assumptions.
- Design a retention targeting rule based on expected value and a fixed campaign budget.
- Run sensitivity analysis over offer cost and assumed probability that an offer saves a would-be churning.

Baseline (Required)

- Logistic regression plus one tree-based model.
- Calibration and threshold analysis.
- Expected-value targeting simulator comparing model-based targeting with simple rules (e.g., highest churn risk only).

Stretch (Optional / Bonus)

- Gradient boosting with SHAP/feature attribution.
- Profit curve / budget optimization across multiple campaign sizes.
- Segment-specific retention policies with fairness or customer-experience discussion.

Success Metrics

- Predictive: PR-AUC/ROC-AUC plus calibration.
- Targeting: precision among contacted customers and coverage of high-risk/high-value customers.
- Business metric: expected net retention value/savings under stated assumptions, with sensitivity ranges.

Deliverables

- Churn scorecard and customer-level targeting table.
- Retention playbook with campaign size, offer assumptions, and expected-value logic.
- Management dashboard with risk, value, target segments, and sensitivity analysis.

Scope Guardrails

- The dataset contains no randomized retention campaign. Do not perform or claim causal uplift modeling from these data.
- The dataset is a teaching/sample dataset; teams must discuss external-validity limitations.
- Assumed save-rate and offer cost must be visible parameters rather than hidden constants.

8. Short-Term Electricity Demand Forecasting & Peak Capacity Planning

Business Question

How accurately can short-term electricity demand be forecast, and how can probabilistic forecasts support peak-capacity planning?

Primary Dataset

- UCI Machine Learning Repository – Power Consumption of Tetouan City. Ten-minute measurements for three distribution zones with temperature, humidity, wind speed, and diffuse-flow variables; no missing values in the published dataset. [UCI – Power Consumption of Tetouan City](#)
- Recommended scope: aggregate to hourly demand and forecast one zone or the combined system; choose a clear horizon such as 1-hour-ahead or day-ahead.

Core Tasks

- Create hourly demand, calendar, lag, rolling, and weather features.
- Establish a strong seasonal-naïve benchmark.
- Forecast demand for a defined short-term horizon using a statistical and/or machine-learning model.
- Identify peak-risk periods and translate forecasts into a capacity-planning decision.
- Produce probabilistic/quantile forecasts or prediction intervals for risk-aware planning.

Baseline (Required)

- Seasonal-naïve baseline plus SARIMAX or LightGBM/XGBoost.
- Rolling or blocked temporal validation.
- Peak-demand analysis with a fixed capacity threshold or reserve-margin scenario.

Stretch (Optional / Bonus)

- Quantile LightGBM/XGBoost or another probabilistic forecasting method.
- Multi-horizon forecasts (e.g., 1, 6, and 24 hours ahead).
- Comparison of zone-specific versus pooled/combined models.

Success Metrics

- Point forecasts: MAE/RMSE and WAPE (or another well-justified scale-aware metric).
- Probabilistic forecasts: pinball loss and empirical interval coverage.
- Peak metric: peak magnitude/timing error and the share of capacity-threshold breaches correctly anticipated.

Deliverables

- Forecasting notebook/pipeline with backtesting results.
- Capacity-planning brief with peak windows, reserve-margin scenarios, and uncertainty interpretation.
- Dashboard showing actual vs. forecast load, forecast intervals, peak risk, and planning scenarios.

Scope Guardrails

- Do not use random train/test splitting for time-series forecasting.
- Deep learning is not required. A well-validated seasonal/statistical or gradient-boosting model is preferred to an under-tuned LSTM.
- Keep the forecast horizon and operational decision explicit throughout the report.

Final Submission Checklist

- The business decision, unit of analysis, target/outcome, and evaluation period are explicitly defined.
- Dataset source, download date, license/terms, and all filtering/joining choices are documented.
- No target leakage is present; all predictors are available at the stated decision time.
- The baseline is complete and compared fairly with any advanced model.
- Validation matches the temporal/spatial structure of the problem.
- Technical metrics and at least one business/decision metric are reported.
- All scenario assumptions are visible, and sensitivity analysis is included where impact is simulated.
- The dashboard supports a concrete decision rather than repeating exploration charts.
- The executive brief states the recommendation, expected impact/range, risks, and limitations in plain business language.
- A third party can rerun the core analysis from the submitted README and code.

Data Sources – Quick Reference

- [Online Retail II \(UCI\)](#)
- [Citi Bike System Data](#)
- [Inside Airbnb – Get the Data](#)
- [Chicago Food Inspections](#)
- [BTS On-Time Performance](#)
- [NYC Motor Vehicle Collisions](#)
- [UK Road Safety Open Data](#)
- [IBM Telco Customer Churn \(Kaggle mirror\)](#)
- [Power Consumption of Tetouan City \(UCI\)](#)

Important: Source availability and terms can change. Verify the source again when the project starts and record the access/download date in the project metadata.